

Minor- I: Probability and Statistics

College For Integrated Studies
University of Hyderabad

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Duration: 60 minutes
Maximum Score: 20 points

Instructor: Dharmendra Kumar
Course Code: MM 212

Instructions: Answers without justification will receive a score of zero.

1. How many vectors $(x_1, \dots, x_k)$ are there for which each x_i is a positive integer such that $1 \leq x_i \leq n$ and $x_1 < x_2 < \dots < x_k$?. [5]
2. Five people, designated as A, B, C, D, E are arranged in linear order. Assuming that each possible order is equally likely, compute the probability that there is exactly one person between A and B . [5]
3. Five balls are randomly chosen without replacement from an urn that contains 5 red, 6 white and 7 blue balls. Find the probability that at least one ball of each color is chosen. [5]
4. Suppose that 20 men at a party throws his hat into the center of the room. The hats are mixed up and then each man randomly selects a hat. what is the probability that none of the men selects his own hat. [5]

All the best !